

Sohum Mehta

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EDUCATION

(incoming) **University of Pennsylvania (UPenn)**, (intended) Cognitive Science & Computer Science (BA), Minor: Statistics and Data Science (Wharton)
Illinois Mathematics and Science Academy (STEM school in America; Residential Life), High School Diploma Expected Graduation: May 2026
GPA: 3.98/4.00 (Unweighted)

Relevant Coursework: Machine Learning, OOP, BC Calculus 1/2/3, Differential Equations, Computational Neuroscience (Independent Study), Biology: Molecular and Cellular, Biology of Behavior, History of Biology and Medicine, Human Anatomy and Physiology

PROFESSIONAL & RESEARCH EXPERIENCES

Explainable AI (XAI) Student Research Intern Remote
Google Quantum AI Lab, Quantum Machine Learning | School-based Internship 06/2025 – Present

- Research variational quantum algorithms (VQE/VQC) and explainable AI (saliency + parameter-shift) under mentorship from Software Lead
- Built reproducible pipelines for hybrid QML experiments with the goal of understanding AI model behavior; documented benchmarks that informed internal research directions.
- Authored work with acceptances including IEEE/ACM AI-ML Systems (Quantum Track), ML4PS Workshop (NeurIPS 2025; sole-author paper), and International Conference for Young Quantum Information Scientists (YQIS) 2025 (poster); additional poster accepted at Purdue’s 2025 Industrial Engineering Quantum AI symposium.

Program Committee Member (Peer-Reviewer) Remote
Association for the Advancement of Artificial Intelligence (AAAI) | Professional Scholarship and Service 07/2025 – 09/2025

- Nominated by AAAI Program Committee Chair for Paper submission(s); assigned 6 AAAI’26 reviews (Main + AI Tracks)
- Reviewed paper of 20+ pages of novel explainable AI research, with structured, standards-aligned feedback.
- Recognized as a respected contributor to rigorous peer review and reproducibility.

Founder and Lead Researcher Remote
Promptonium 06/2025 – 09/2025

- Launched a research-driven platform that “genotypes” prompts to improve LLM efficacy, effectively mapping human intention to LLM outputs
- Microsoft-supported (\$1K); first-author paper accepted to EMNLP BlackBox 2025 describing the novel *Prompt Genotyping* framework (paper cited by Carnegie Mellon University and Microsoft).
- Launched an educational AI prompting tool integrating this research framework, allowing researchers to learn prompt engineering; used by PhDs.

Internal Project Fellow (Software Engineering and Research) Remote
Wedge (YC S25), Backed by Y-Combinator & Founders from Stanford Medicine,, NVIDIA, Intel, Rivian, and SoFi | Technical Consultant 08/2025 – 08/2025

- Selected for a YC-backed fellowship (top <1%) as the only high-school operator, collaborating with undergraduate and graduate teams.
- Led technical + strategy work in Edge Consulting (applied AI, workflow automation, and analytics-to-action) for major hospital systems; bridged prototypes to client-ready solutions with SWE/research teams.
- Scoped and delivered projects in a model where peers earned up to \$180/hr, validating market demand and exposure to healthcare innovation.

Independent Study in NeuroAI (Computational Neuroscience) and Independent Research in Medical AI (Informatics) Aurora, IL
Illinois Mathematics and Science Academy 09/2024 – 10/2025

- Studied neuroscience through structured course with practical research applications through modeling action potentials and neuronal excitability; Invited to EAMH Brain Informatics (Springer) workshop as an Oral Presenter (youngest amongst PhDs/Professors; full-paid registration) and Sigma Xi IFoRE (intl.)
- Led neuroscience-inspired ML research on research drawing on associative memory and Hopfield-network theory; First-authored paper accepted to ICLR NFAM 2026
- Selected (by school) to showcase my AI work to legislators and corporate partners; received AI/Research certificate from Rep. Regan Deering
- First-authored behavioral AI theft detection research, securing medical models via uncertainty signatures with zero accuracy degradation; Publication at Lock-LLM Workshop (NeurIPS 2025)

Bioinformatics Research Intern, Winter Lab Hybrid
Northwestern University (Feinberg), Computational Biology | Internship 06/2025 – 09/2025

- Engineered a reusable single-cell RNA-seq web app for lab-wide analysis and visualization of immune-cell gene programs.
- Led biomarker discovery across inflammatory disease datasets; collaborated with immunologists to validate signals.
- First-author paper Published at ACM-BCB 2025 (Proceedings, poster presentation)

Computational Neurolinguistics Student Researcher Remote
Carnegie Mellon University 09/2024 – 05/2025

- Modeled lexical retrieval using EEG and reaction-time data; led analysis of speech-pathway dynamics across picture vs. word processing.
- National JSHS Finalist (top ~3%; \$1000); accepted for Oral Presentation at Springer Brain Informatics (specialized international conference)
- Produced Python notebooks and regression workflows that generalized across datasets and stimuli types.

LEADERSHIP & EXTRACURRICULAR INVOLVEMENT

Co-Founder & Co-President <i>Pay it Forward (PiF) Science, 501(c)(3)</i> - Democratized STEM access for 3,000+ students through weekly workshops and learning tools. - Recruited and led ~70 advisors; built 9 international NGO partnerships to expand programming. - Sustained support by Google Ads Grant (~\$120K/yr), scaling multi-site program delivery and campaigns.	Global 11/2023 – Present
Independent Study in Jainism <i>Arihanta Institute</i> - Awarded full term scholarship; among the youngest learners in a world-class graduate school program on Jain ethics and philosophy. - Integrated philosophical frameworks (truth-seeking, non-attachment) into research leadership and team culture.	Remote 02/2025 - Present
Mentor and Teaching Facilitator <i>Leadership Education and Development (LEAD) Program Illinois Mathematics and Science Academy</i> - Selective leadership program; teaching a weekly leadership seminar to ~15 underclassmen each year in communication, team dynamics, and project ownership in classroom settings - Designed lesson plans and activities; ran feedback loops to improve engagement and measurable skill adoption. - Contributed 200+ service hours facilitating instruction, mentoring, and program operations.	Aurora, IL 08/2023 - Present
International Chess Player <i>Chess Club</i> - Former top-100 U.S. youth player; internationally ranked (top 5% by FIDE); Board 1 at former school's Chess Club - Led daily team training and coached elementary students in competitive chess (paid coaching role). - Applied deliberate practice and game review frameworks to improve team performance.	Hybrid 2015 – Present

AWARDS & HONORS

NeurIPS 2025 — Publication and Poster Presentations (2x): ML4PS (sole-author paper) and Lock LLM Workshop (peer-reviewed proceedings, Springer).

2025 National JSHS Finalist — Prestigious science fair, Selected as 249 from 8,000+ researchers (<3%). Advanced via oral presentation; \$1,000 scholarship

ACM-BCB 2025 — 2x First-author poster papers Published in proceedings (flagship computational biology venue)

EMNLP 2025 — First-author paper accepted to BlackBoxNLP; the research framework “Prompt Genotyping”

ICLR 2026 — First-author paper accepted to New Frontiers in Associative Memories (NFAM)

Brain Informatics (Springer) 2025 — 2x Accepted Oral Presentation in computational neuro research + full-paid Workshop invite

AAAI-26 Program Committee Member (Main & AI Track) — Nominated by PC Chair; rare HS reviewer among PhDs/faculty

YC-Sponsored Wedge Fellowship (S25) — 0.7% admit; only high-school operator among undergrad/grad peers

Congressional App Challenge Winner (IL-16, 2024) — Rep. Darin LaHood; coded BiteSense, an AI diagnostics app for bug-bites

IEEE/ACM AI-ML Systems 2025 (Quantum Track) — Two first-author papers accepted.

National Security Agency's (NSA) ChiCyber Conference (2025) — Youngest invited presenter (NTFS forensics research)

Certificate of Excellence in AI & Research — Rep. Regan Deering, after international recognition

2024 YICU Service Award Winner — Selected as 1 of 10 from 50 Nominees, received \$250 for a local nonprofit

SKILLS

Technical/Design: Python (many libraries), Prompt Engineering, Java, R (some), Google Cirq, MATLAB, Microsoft Office 365

Research: Artificial Intelligence, Computational Neuroscience, Psycholinguistics, Natural Language Processing, Quantum Machine Learning, Bioinformatics

Soft: Communication effectiveness, Willingness to Learn, Confidence, Mentorship, Purpose-oriented, Adaptability, Moral Integrity

Languages: English (Fluent), Marathi (Conversationally Fluent)

Additional Links - Google Scholar: <https://scholar.google.com/citations?user=ZtOb364AAAAJ&hl=en>

Summary Statement:

Interdisciplinary at heart: linguistics, philosophy, neuroscience, computer science, and AI.

My goal? Understanding the mind and body, specifically how we think or produce intelligence. I use computation as a tool to come closer to those questions.

All my experiences share this line of reasoning, whether it's computational/AI or behavioral modelling.